SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

E – Commerce Platform for Personalized Retail Experiences

Group No: 25-26J-526

Checklist 02

30/04/2026

SmartRetailX Deployment Report

Project: Smart Retail X

Report Type: Production Deployment Report

Date: April 30, 2026

Environment: Kubernetes Production (Namespace: smart-retail-x)

Deployment Details

- Deployment package: Kubernetes manifests under k8s/prod
- Target version/tag: latest (container images)
- Deployment date/time: April 30, 2026 (UTC)
- Target environment: Production of Kubernetes cluster

Scope of Changes

- Added production Kubernetes namespace, configuration, and secret template
- Added infrastructure manifests for PostgreSQL and RabbitMQ with persistent storage
- Added application deployments/services for API, web, core, BI, WebSocket, ML, promotion, segmentation, agent, and SinLlama services
- Added ingress routing for app, API, and BI endpoints
- Added Kustomize entry point and deployment instructions

Deployment Process

- Prepared manifests in k8s/prod:

  namespace.yaml, configmap.yaml, secret.example.yaml, infra.yaml, apps.yaml, ingress.yaml, kustomization.yaml

- Created deployment guide in k8s/prod/README.md
- Updated main README production section with Kubernetes instructions
- Expected deployment command: kubectl apply -k k8s/prod

Status and Results

- Deployment preparation status: Success (manifests and docs completed)
- Runtime cluster deployment status: Pending execution in target cluster
- Local validation status: Partial (static file creation complete; cluster tooling unavailable locally)

Issues Encountered

- kubectl and kustomize binaries were not installed in the current environment
- As a result, render/apply validation could not be executed locally
- Rollback: Not applicable (no cluster applied performed from this environment)

Post-Deployment Validation Plan

After applying to cluster, run:

- kubectl get pods -n smart-retail-x
- kubectl get svc -n smart-retail-x
- kubectl get ingress -n smart-retail-x
- Verify health endpoints:

/api/health/live (api-gateway), /health (agent/ml/promotion/sinlama), /docs (segmentation)

Responsible Parties

- Prepared by: Engineering (Codex session)
- Deployment executor: DevOps/Platform Engineer (cluster access required)
- Approver: Project technical lead / release manager

Notes

The report records deployment readiness and implementation of Kubernetes production artifacts for Smart Retail X. Final production success confirmation requires cluster-side apply and verification.

Appendices

```
 ! image smart-retail-x/promotion-engine:latest          pull access denied for smart-retail-x/promotion-engine, repository does not exist or may require 'docker login'   1.3s
[+] Building 11.9s (17/68)
 => [internal] load local bake definitions                                                                                                                                  0.0s
 => => reading from stdin 4.23kB                                                                                                                                            0.0s
 => [websocket-service internal] load build definition from Dockerfile                                                                                                      0.0s
 => => transferring dockerfile: 1.37kB                                                                                                                                      0.0s
 => [promotion-engine internal] load build definition from Dockerfile                                                                                                       0.1s
 => => transferring dockerfile: 780B                                                                                                                                        0.0s
 => [bi-dashboard-ml-service internal] load build definition from Dockerfile                                                                                                0.1s
 => => transferring dockerfile: 729B                                                                                                                                        0.0s
 => [api-gateway internal] load build definition from Dockerfile                                                                                                            0.1s
 => => transferring dockerfile: 1.34kB                                                                                                                                      0.0s
 => [segmentation-service internal] load build definition from Dockerfile                                                                                                   0.1s
 => => transferring dockerfile: 1.00kB                                                                                                                                      0.0s
 => [bi-dashboard-services-gateway internal] load build definition from Dockerfile                                                                                          0.1s
 => => transferring dockerfile: 1.42kB                                                                                                                                      0.0s
 => [core-service internal] load build definition from Dockerfile                                                                                                           0.1s
 => => transferring dockerfile: 1.35kB                                                                                                                                      0.0s
 => [web internal] load build definition from Dockerfile                                                                                                                    0.1s
 => => transferring dockerfile: 1.21kB                                                                                                                                      0.0s
 => [core-service internal] load metadata for docker.io/library/node:20-bookworm-slim                                                                                       2.6s
 => [segmentation-service internal] load metadata for docker.io/library/python:3.11-slim                                                                                    2.8s
 => [web internal] load metadata for docker.io/library/nginx:1.27-alpine                                                                                                    3.5s
 => [web internal] load .dockerignore                                                                                                                                       0.0s
 => => transferring context: 406B                                                                                                                                           0.0s
 => [web internal] load build context                                                                                                                                       8.9s
 => => transferring context: 80.11MB                                                                                                                                        8.9s
 => [api-gateway base 1/3] FROM docker.io/library/node:20-bookworm-slim@sha256:2cf067cfed83d5ea958367df9f966191a942351a2df77d6f0193e162b5febfc0                                8.9s
 => => resolve docker.io/library/node:20-bookworm-slim@sha256:2cf067cfed83d5ea958367df9f966191a942351a2df77d6f0193e162b5febfc0                                                0.0s
 => => sha256:804d4d68057cbb26cbcde9a735148ebc6589911bad32cf9dbddb5b0ba878bf1f 41.42MB / 41.42MB                                                                            2.7s
 => => sha256:64cfb949317c9824fd081d55c59a7225bfe1f650c6395ba42d7ed0ccf51993d2 1.71MB / 1.71MB                                                                              0.6s
 => => sha256:3c02fd806613748139b8b8a14a69ae3518a1b0e863c57c35aada3b92ed87fb31 447B / 447B                                                                                  0.9s
 => => sha256:e54aec64c365815cd7b91e718f00ac6e625562b1de09036b38614621b42c7582 3.31kB / 3.31kB                                                                              1.0s
 => => sha256:ff86ea2e5edce334d19a34fbc65d1a511aa1fc823dba1110422f991aa56b44d4 28.24MB / 28.24MB                                                                            1.8s
 => => extracting sha256:ff86ea2e5edce334d19a34fbc65d1a511aa1fc823dba1110422f991aa56b44d4                                                                                   5.3s
 => => extracting sha256:e54aec64c365815cd7b91e718f00ac6e625562b1de09036b38614621b42c7582                                                                                   0.1s
 => => extracting sha256:804d4d68057cbb26cbcde9a735148ebc6589911bad32cf9dbddb5b0ba878bf1f                                                                                   0.9s
 => [bi-dashboard-ml-service internal] load .dockerignore                                                                                                                   0.1s
 => => transferring context: 2B                                                                                                                                             0.0s
```

```
 => [bi-dashboard-ml-service 2/8] WORKDIR /app                                                                                                                              0.1s
 => [bi-dashboard-ml-service 3/8] RUN apt-get update && apt-get install -y     gcc     g++     libpq-dev     && rm -rf /var/lib/apt/lists/*                          5.5s
 => => # Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
 => => # Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
 => => # Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
 => => # Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [129 kB]
 => => # Fetched 9895 kB in 5s (2197 kB/s)
 => => # Reading package lists...
 => [segmentation-service 3/8] RUN apt-get update     && apt-get install -y --no-install-recommends build-essential curl ffmpeg libpq-dev     && rm -rf /var/lib/apt/lists/*   5.5s
 => => # Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
 => => # Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
 => => # Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
 => => # Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [129 kB]
 => => # Fetched 9895 kB in 4s (2371 kB/s)
 => => # Reading package lists...
 => [websocket-service runtime 2/5] WORKDIR /app                                                                                                                            0.1s
 => [web base 2/3] WORKDIR /workspace                                                                                                                                       0.1s
 => [api-gateway runtime 3/5] RUN apt-get update     && apt-get install -y --no-install-recommends ca-certificates curl openssl     && rm -rf /var/lib/apt/lists/*          4.6s
 => => # Get:1 http://deb.debian.org/debian bookworm InRelease [151 kB]
 => => # Get:2 http://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
 => => # Get:3 http://deb.debian.org/debian-security bookworm-security InRelease [48.0 kB]
 => => # Get:4 http://deb.debian.org/debian bookworm/main amd64 Packages [8792 kB]
 => => # Get:5 http://deb.debian.org/debian bookworm-updates/main amd64 Packages [6924 B]
 => => # Get:6 http://deb.debian.org/debian-security bookworm-security/main amd64 Packages [297 kB]
 => [websocket-service base 3/3] RUN apt-get update     && apt-get install -y --no-install-recommends ca-certificates openssl     && rm -rf /var/lib/apt/lists/*     && corepack enable     && corepack p  4.6s
 => => # Get:1 http://deb.debian.org/debian bookworm InRelease [151 kB]
 => => # Get:2 http://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
 => => # Get:3 http://deb.debian.org/debian-security bookworm-security InRelease [48.0 kB]
 => => # Get:4 http://deb.debian.org/debian bookworm/main amd64 Packages [8792 kB]
 => => # Get:5 http://deb.debian.org/debian bookworm-updates/main amd64 Packages [6924 B]
 => => # Get:6 http://deb.debian.org/debian-security bookworm-security/main amd64 Packages [297 kB]
 => [web builder 3/9] RUN corepack enable     && corepack prepare pnpm@10.32.1 --activate                                                                                   4.6s
 => => # Preparing pnpm@10.32.1 for immediate activation...
```

```
=> [web builder 6/9] COPY libs/shared-types/package.json libs/shared-types/package.json                                   0.0s
=> [web builder 7/9] RUN pnpm install --frozen-lockfile                                                                  201.7s
=> => # Progress: resolved 2640, reused 0, downloaded 2632, added 840
=> => # Progress: resolved 2640, reused 0, downloaded 2632, added 929
=> => # Progress: resolved 2640, reused 0, downloaded 2632, added 1001
=> => # Progress: resolved 2640, reused 0, downloaded 2632, added 1102
=> => # Progress: resolved 2640, reused 0, downloaded 2632, added 1163
=> => # Progress: resolved 2640, reused 0, downloaded 2632, added 1219
=> [api-gateway deps 1/4] COPY package.json pnpm-lock.yaml pnpm-workspace.yaml nx.json tsconfig.base.json ./              0.3s
=> [websocket-service deps 2/4] COPY apps/web/package.json apps/web/package.json                                          0.1s
=> [websocket-service deps 3/4] COPY libs/shared-types/package.json libs/shared-types/package.json                       0.1s
=> [websocket-service deps 4/4] RUN pnpm install --frozen-lockfile                                                      174.4s
=> => # Progress: resolved 2640, reused 0, downloaded 2373, added 706
=> => # Progress: resolved 2640, reused 0, downloaded 2387, added 710
=> => # Progress: resolved 2640, reused 0, downloaded 2401, added 714
=> => # Progress: resolved 2640, reused 0, downloaded 2430, added 722
=> => # Progress: resolved 2640, reused 0, downloaded 2445, added 727
=> => # Progress: resolved 2640, reused 0, downloaded 2457, added 730
=> [bi-dashboard-ml-service 4/8] COPY requirements.txt .                                                                  0.1s
=> [bi-dashboard-ml-service 5/8] RUN pip install --no-cache-dir -r requirements.txt                                      157.8s
=> => # Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
=> => # Downloading stanio-0.5.1-py3-none-any.whl (8.1 kB)
=> => # Installing collected packages: websockets, uvloop, typing-extensions, tqdm, threadpoolctl, slicer, six, pyyaml, python-multipart, python-dotenv, pyparsing, psycopg2-binary, pillow, packaging, nvidia
=> => # -nccl-cu12, numpy, loguru, llvmlite, kiwisolver, joblib, importlib_resources, idna, httptools, h11, greenlet, fonttools, cycler, cloudpickle, click, certifi, annotated-types, annotated-doc, uvicorn,
=> => #  typing-inspection, stanio, sqlalchemy, scipy, python-dateutil, pydantic-core, numba, httpcore, contourpy, anyio, xgboost, watchfiles, starlette, scikit-learn, pydantic, pandas, matplotlib, httpx, h
=> => # olidays, shap, fastapi, cmdstanpy, prophet
=> [promotion-engine 4/7] COPY apps/personalized-promotion-engine-ml-service/requirements.txt /tmp/requirements.txt        0.4s
=> [segmentation-service 4/8] COPY apps/segmentation-service/src/requirements.txt /tmp/requirements.txt                   0.4s
=> [promotion-engine 5/7] RUN python -m pip install --upgrade pip     && pip install --no-cache-dir -r /tmp/requirements.txt  56.9s
=> => #     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.8/3.8 MB 128.2 MB/s  0:00:00
=> => # Downloading watchfiles-1.1.1-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (456 kB)
=> => # Downloading websockets-16.0-cp311-cp311-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (184 kB)
=> => # Installing collected packages: websockets, uvloop, typing-extensions, tqdm, threadpoolctl, six, pyyaml, python-dotenv, pyparsing, psycopg2-binary, pillow, packaging, numpy, kiwisolver, joblib, idna,
=> => #  httptools, h11, fonttools, faker, cycler, click, annotated-types, annotated-doc, uvicorn, typing-inspection, scipy, python-dateutil, pydantic-core, contourpy, anyio, watchfiles, starlette, scikit-l
=> => # earn, pydantic, pandas, matplotlib, seaborn, fastapi
=> [segmentation-service 5/8] RUN python -c "from pathlib import Path; p=Path('/tmp/requirements.txt'); b=p.read_bytes(); p.write_text((b.decode('utf-16') if b[:2] in (b'\xff\xfe', b'\xfe\xff') or b'  56.9s
=> => # Downloading uvloop-0.22.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (3.8 MB)
=> => #     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.8/3.8 MB 138.4 MB/s  0:00:00
=> => # Installing collected packages: pytz, websockets, uvloop, urllib3, tzdata, typing_extensions, threadpoolctl, six, shellingham, rignore, PyYAML, python-multipart, python-dotenv, Pygments, psycopg2-bin
=> => # ary, numpy, mdurl, MarkupSafe, joblib, idna, httptools, h11, greenlet, fastar, dnspython, colorama, click, certifi, annotated-types, annotated-doc, uvicorn, typing-inspection, SQLAlchemy, sentry-sdk
=> => # , scipy, python-dateutil, pydantic_core, markdown-it-py, Jinja2, httpcore, email-validator, anyio, watchfiles, starlette, scikit-learn, rich, pydantic, pandas, httpx, typer, rich-toolkit, fastapi, f
=> => # astapi-cloud-cli, fastapi-cli
```